RoBERTa-based Sentiment Classification with Ensemble Approach

Fine-tuning a state-of-the-art language model for 3-class sentiment analysis — with multi-seed training and ensemble methods to maximize accuracy and robustness.

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Project Introduction

Base Model

RoBERTa-base — fine-tuned for 3-class(**Negative**/**Neutral**/**Positive**.) Sentiment Analysis on English text .

Hyperparameters

Experiment Design

Trained with **3 random seeds** (42, 43, 44). Best model selected by validation **macro-F then doing the ensamble learning.**

Epochs	Learning Rate	Batch	Decay
5	2e-5	16	0.01

Data Sources & Overview

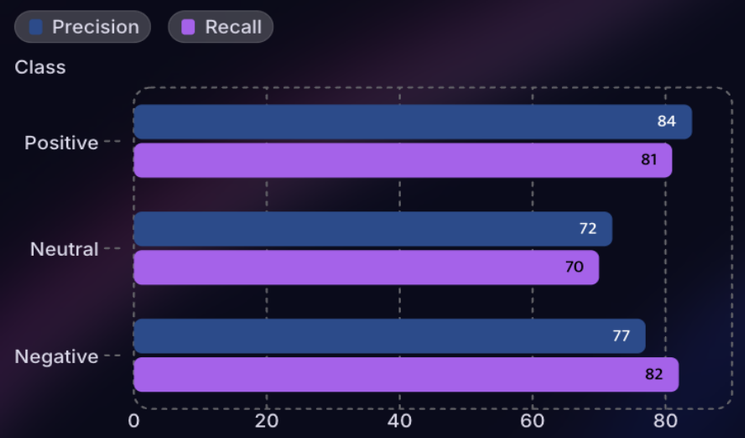
Our project leverages three publicly available datasets, carefully selected to provide a rich and diverse foundation for sentiment classification. Each dataset contributes unique conversational contexts and emotion annotations, crucial for training a robust RoBERTa model.

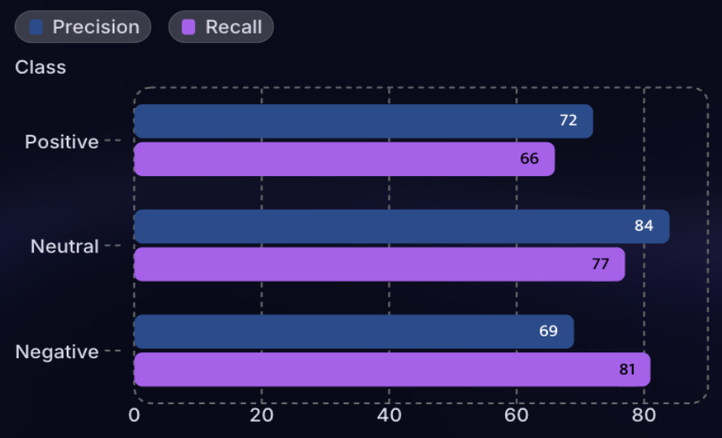
Dataset Name	Type	Key Fields	Size (samples)
EmpatheticDialogues	API	Utterance, Context, Self-evaluation	10,000
DailyDialog	API	Dialogue, Emotion, Act	10,000
go_emotions	API	Text, Labels (27 emotions)	10,000
Mixed_dataset	combined	mix with the datasets above	10,000

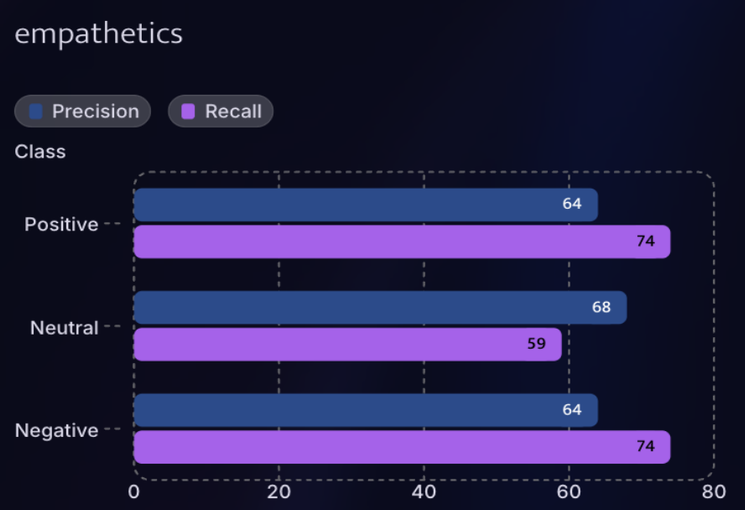
Collectively, these datasets offer over 40,000 conversational turns and texts, annotated with various emotional states, which will be mapped to our 3-class sentiment target (positive, neutral, negative). In addition, the dataset was split into train(80%) / validation(10%) / test(10%).

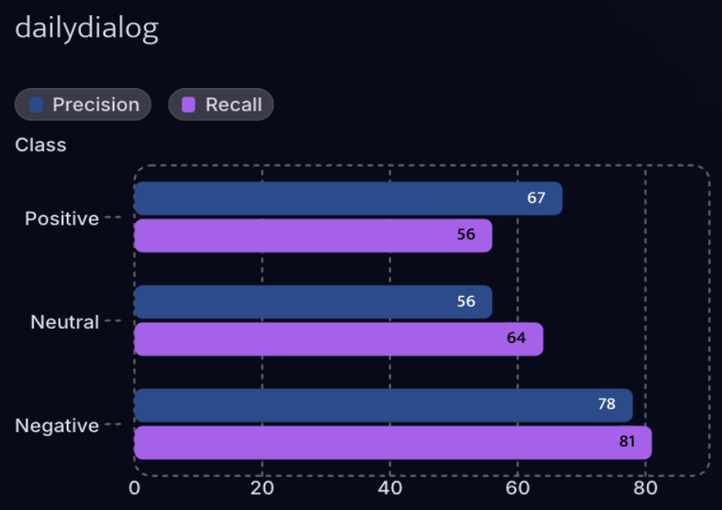
Evaluation & Analysis

Categorized by datasets



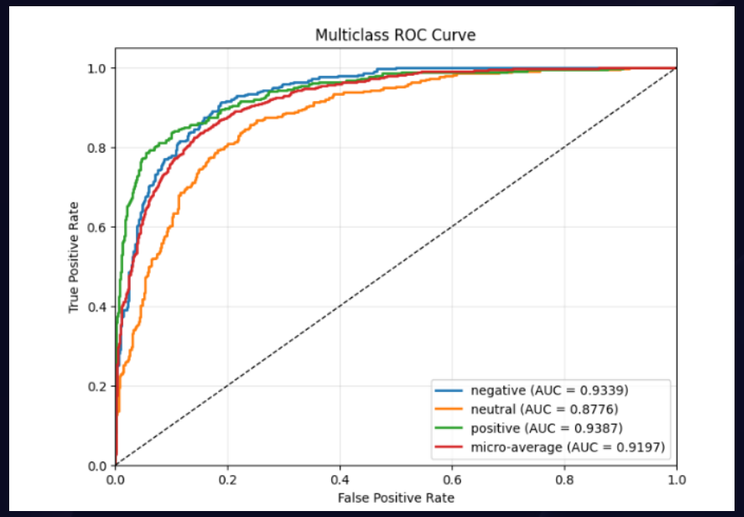




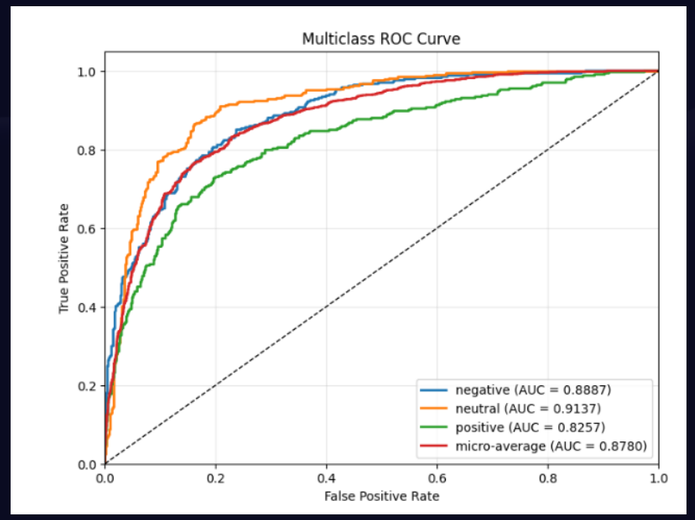


Evaluation & Analysis

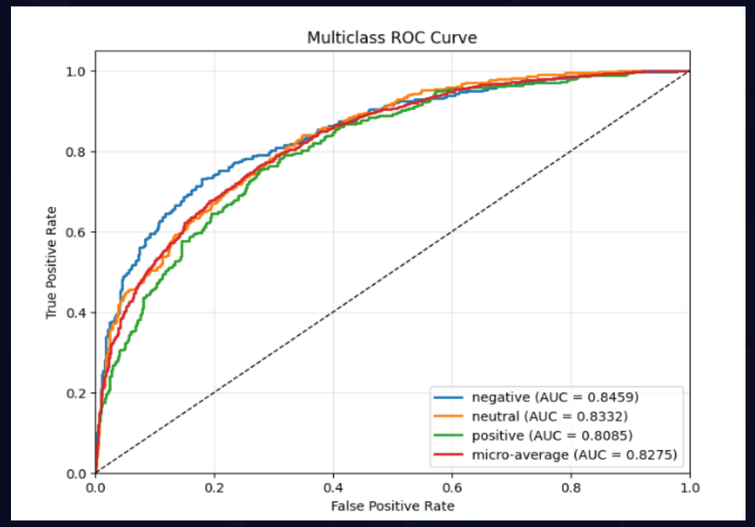
go_emotions



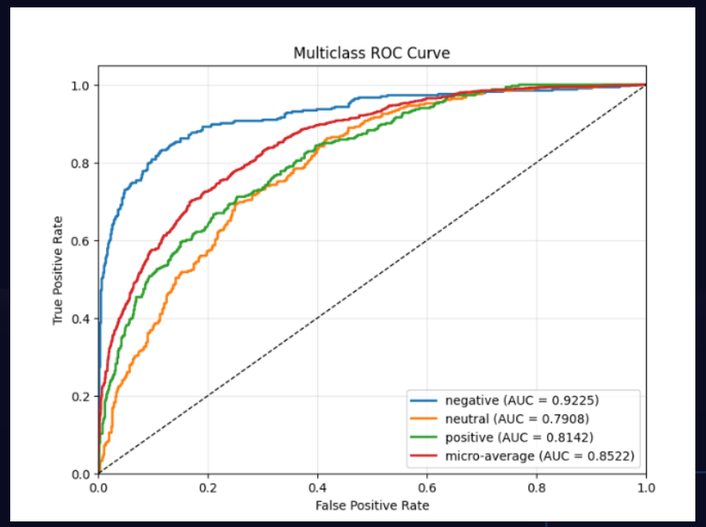
mixed_data



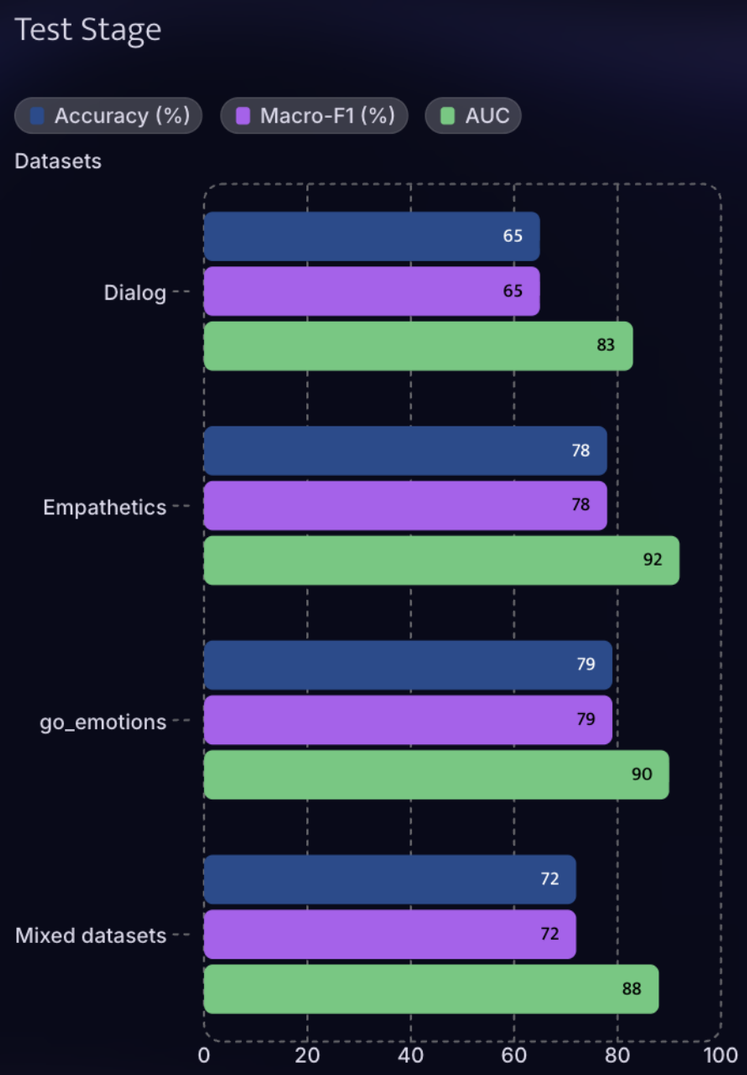
empathetics



dailydialog



Evaluation & Analysis



Conclusion

Current Discover

1 Multi-Seed Ensemble Robustness

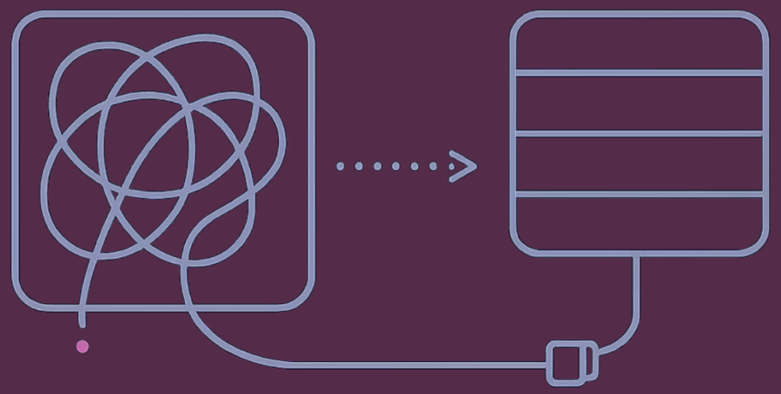
Training across multiple seeds (e.g., 42, 43, 44) reduces variance caused by random initialization and stochastic training dynamics. Compared to single-seed models, ensemble predictions provide more stable and consistent performance, demonstrating improved robustness.

2 High Generalization

The model shows a small performance gap between validation and test sets, suggesting minimal overfitting.
This indicates that the model successfully captures generalizable patterns and can transfer well across similar datasets. The consistent performance across multiple datasets further supports the model's robustness and adaptability.

3 Learning Pattern Bias between neutral & positive

AUC scores are significantly higher than F1 scores, indicating that the model can effectively rank class probabilities but struggles with final classification decisions.This gap reveals that decision boundaries between classes—especially **Neutral and Positive**—are not well defined. As a result, the model frequently misclassifies samples between these semantically similar categories.



Challenges & Limitations

Dataset Quality Enhance

Results on the *go_emotions* dataset are significantly better than those on *DailyDialog* and *Empathetics*. This suggests that performance is highly dependent on dataset quality and clarity. Datasets with clearer emotional signals are easier for the model to learn, while conversational datasets with ambiguous sentiment introduce challenges.

Computing Resource Limitation.

The model may not fully capture complex emotional nuances.
A simple classification head limits its ability to model subtle sentiment variations. Training larger models requires significant GPU resources. Limited computation restricts exploration of higher-capacity models.